

Reinforcement Learning for UAV Attitude Control

Moving Beyond
Proportional-Integral-
Derivative (PID) Systems



The Rigid Reality of PID Controllers

Proportional-Integral-Derivative (PID) controllers are the dominant standard for UAV autopilots, excelling in stable, predictable environments.

However, they are inherently reactive. Exposed to unknown dynamics—like wind gusts or variable payloads—a PID controller's fixed linear parameters become brittle and suboptimal.

Ideal Conditions



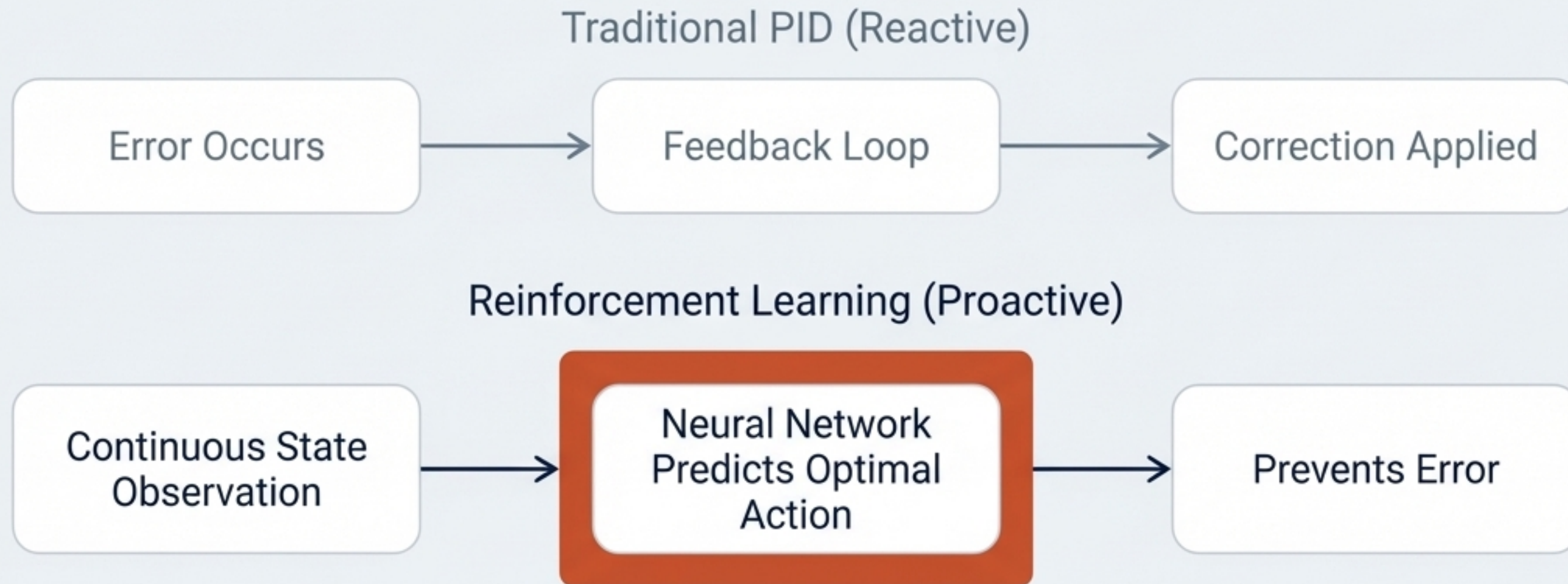
Unpredictable Dynamics



Transitioning from Reactive Feedback to Adaptive Intelligence

Artificial Neural Networks trained via Reinforcement Learning can learn highly accurate offline models of aircraft dynamics and adapt in real-time.

Unlike PID's reactive feedback loops, neural network control is proactive. It anticipates and outputs corrective control signals before a catastrophic error occurs.



The Three Open Challenges in RL Flight Control

C1: Precision & Accuracy

Can RL handle the zero-tolerance, time-critical demands of actual attitude control without causing instability?

Core Focus of Research

C2: Robustness & Adaptation

What is the safe operating range before online adaptation is required due to hardware reality gaps?

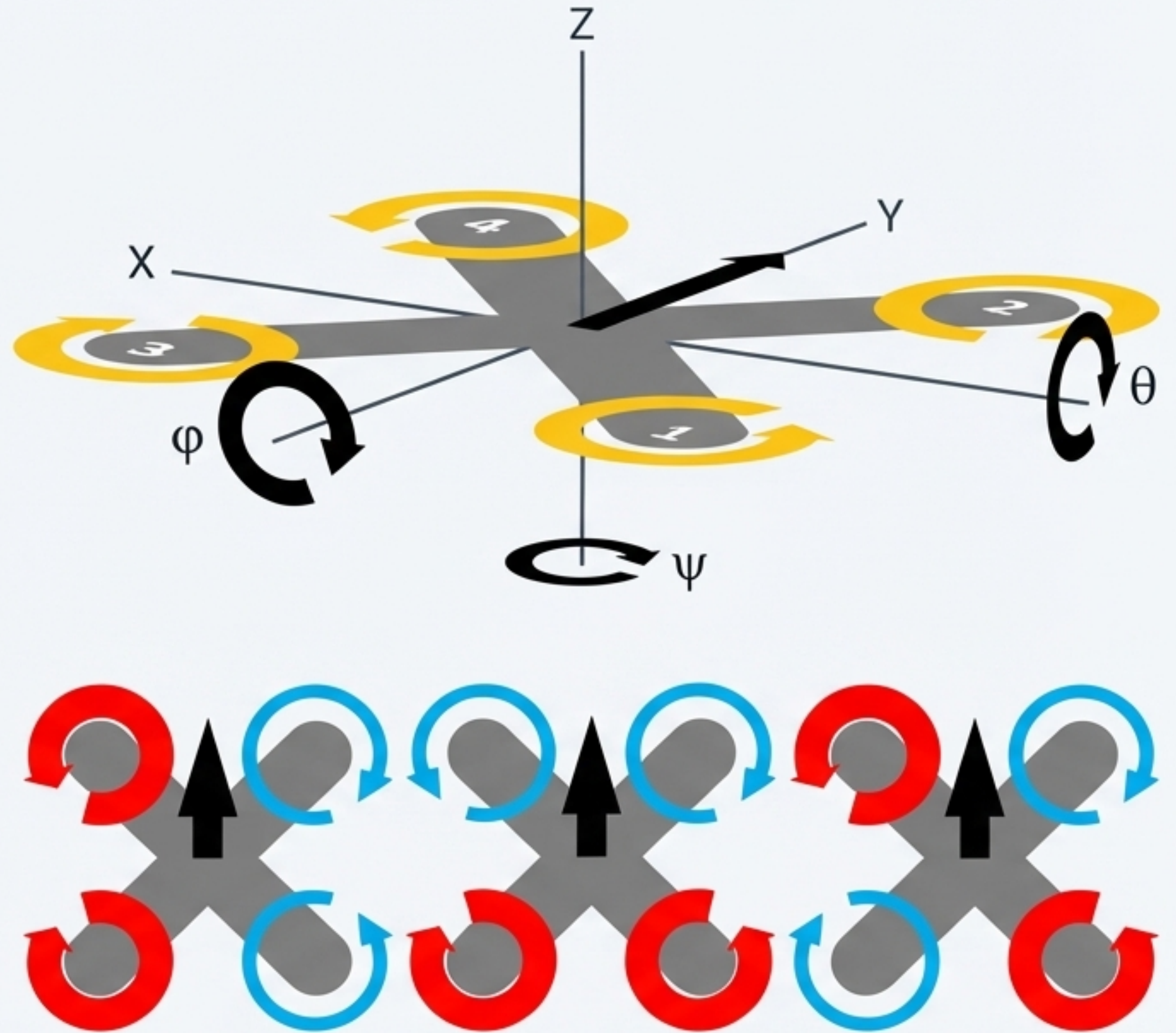
C3: Reward Engineering

How do we design reward signals for complex goals like minimizing energy consumption or surviving physical damage?

The Challenge of the Inner Loop

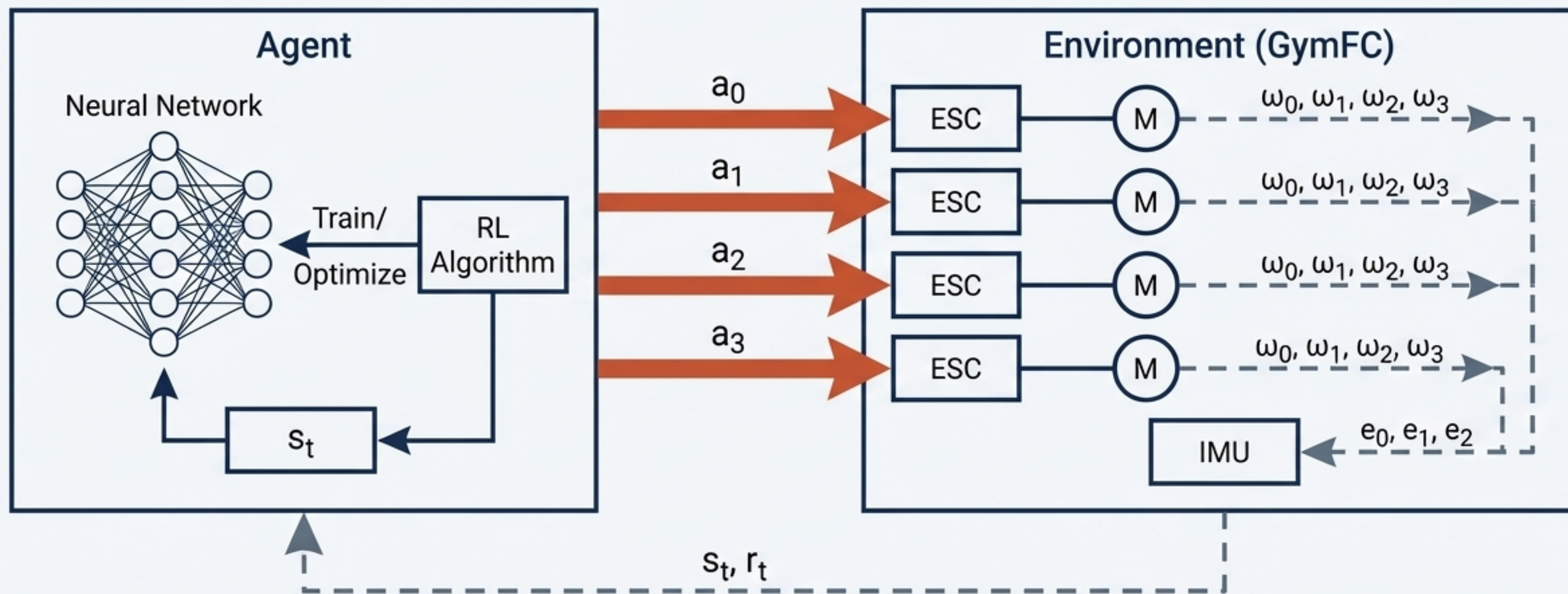
UAV control is split. The **Outer Loop** handles slow, mission-level navigation. The **Inner Loop** handles split-second attitude control (Roll, Pitch, Yaw).

Inner loop control dictates specific angular velocity to maintain six degrees of freedom. The slightest deviation leads to catastrophic failure.



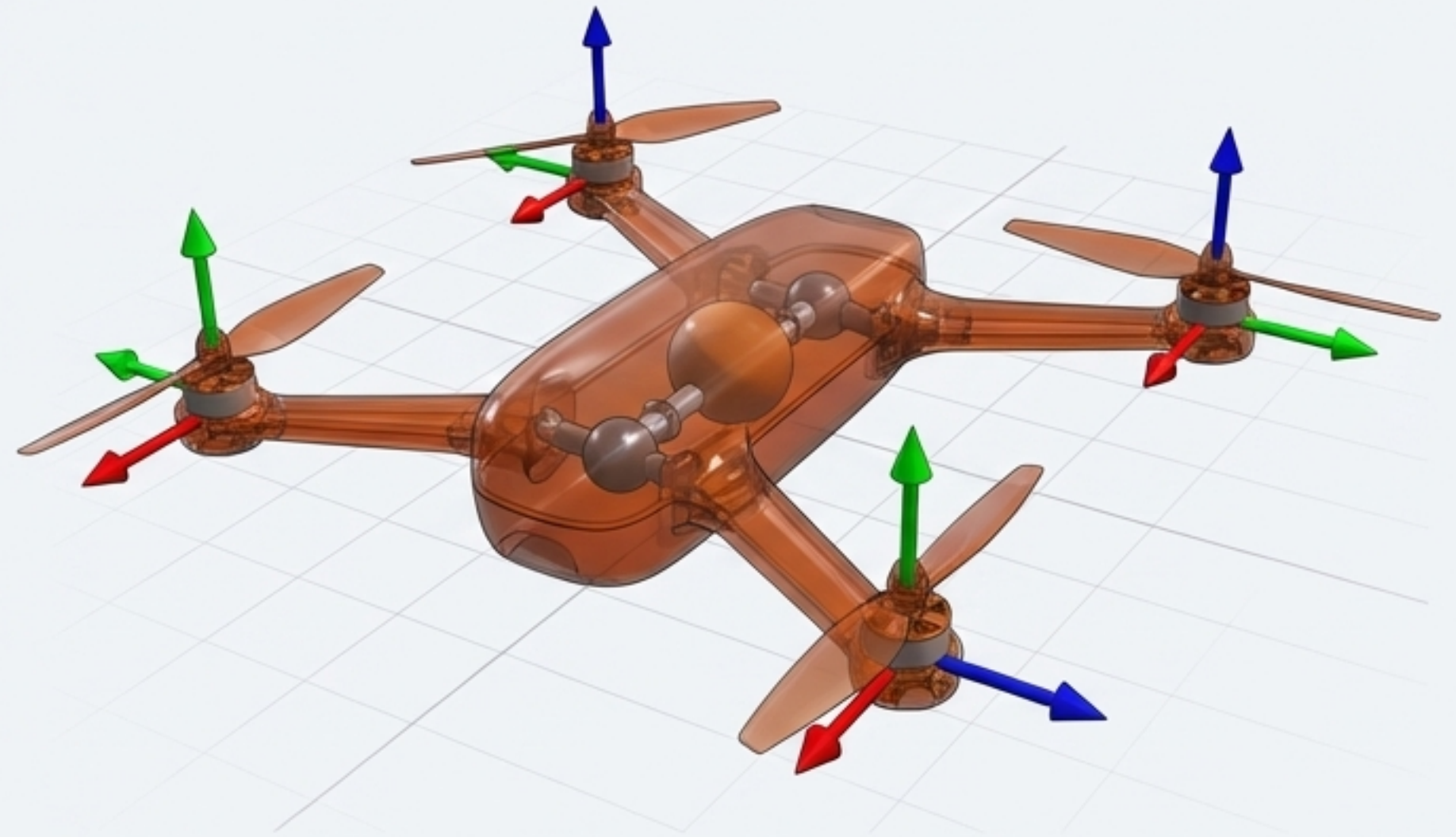
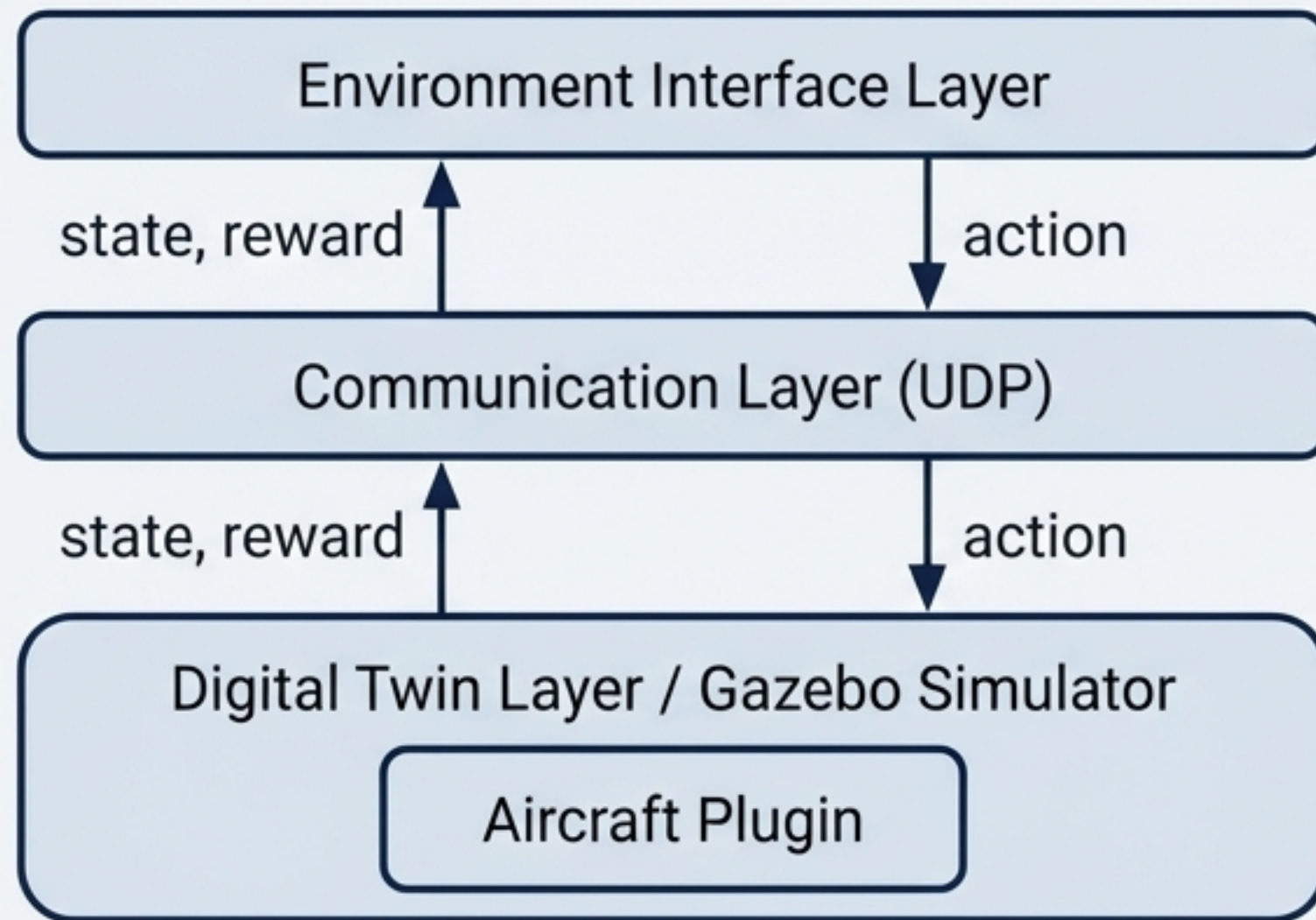
Introducing GymFC: The Open-Source Training Arena

To safely test RL for the inner loop, researchers built GymFC—a novel OpenAI Gym environment utilizing a high-fidelity Gazebo physics simulator. Operating in continuous action and observation spaces, the agent receives angular velocity errors and directly outputs four continuous control signals.



High-Fidelity Digital Twinning Closes the Reality Gap

- GymFC operates on a digital replica of an Iris quadcopter, exposing identical interfaces to actuators and sensors as physical hardware.
- Direct motor updates bypass simulator bottlenecks, driving the simulation clock to achieve a 1kHz physics evolution (1,000 steps per second).



The Contenders: State-of-the-Art RL Algorithms

Three neural network architectures capable of continuous action control were selected to pilot the digital twin.



DDPG

Deep Deterministic Policy Gradient

An actor-critic, model-free algorithm.



TRPO

Trust Region Policy Optimization

A natural gradient policy method guaranteeing monotonic improvements.



PPO

Proximal Policy Optimization

A leading policy gradient method known for outperforming others in highly challenging environments.

The Training Regimen: 10 Million Simulation Steps

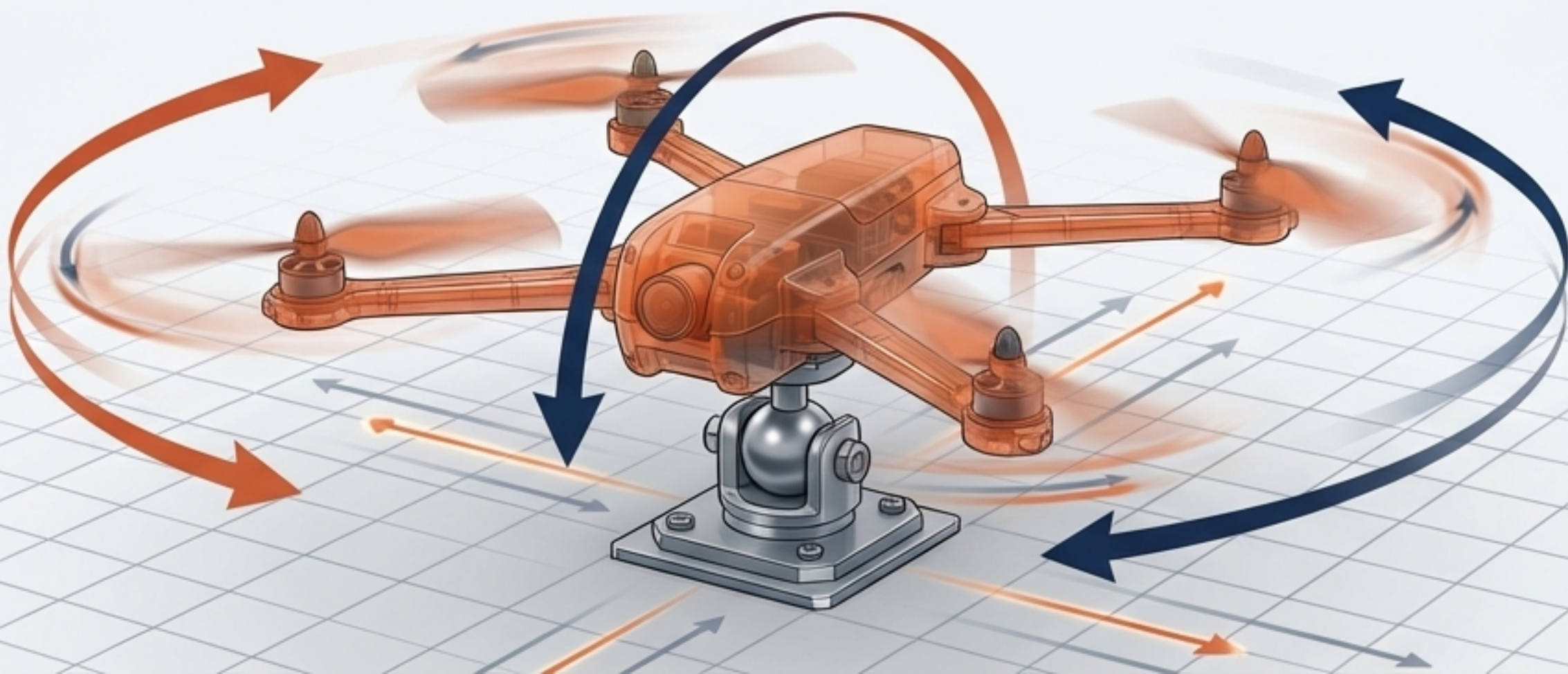
Each agent was subjected to extreme episodic tasks from a dead rest. The singular goal: minimize the angular velocity error against randomly sampled setpoints up to ± 300 degrees/second.

10,000

Extreme Episodic Tasks

10,000,000

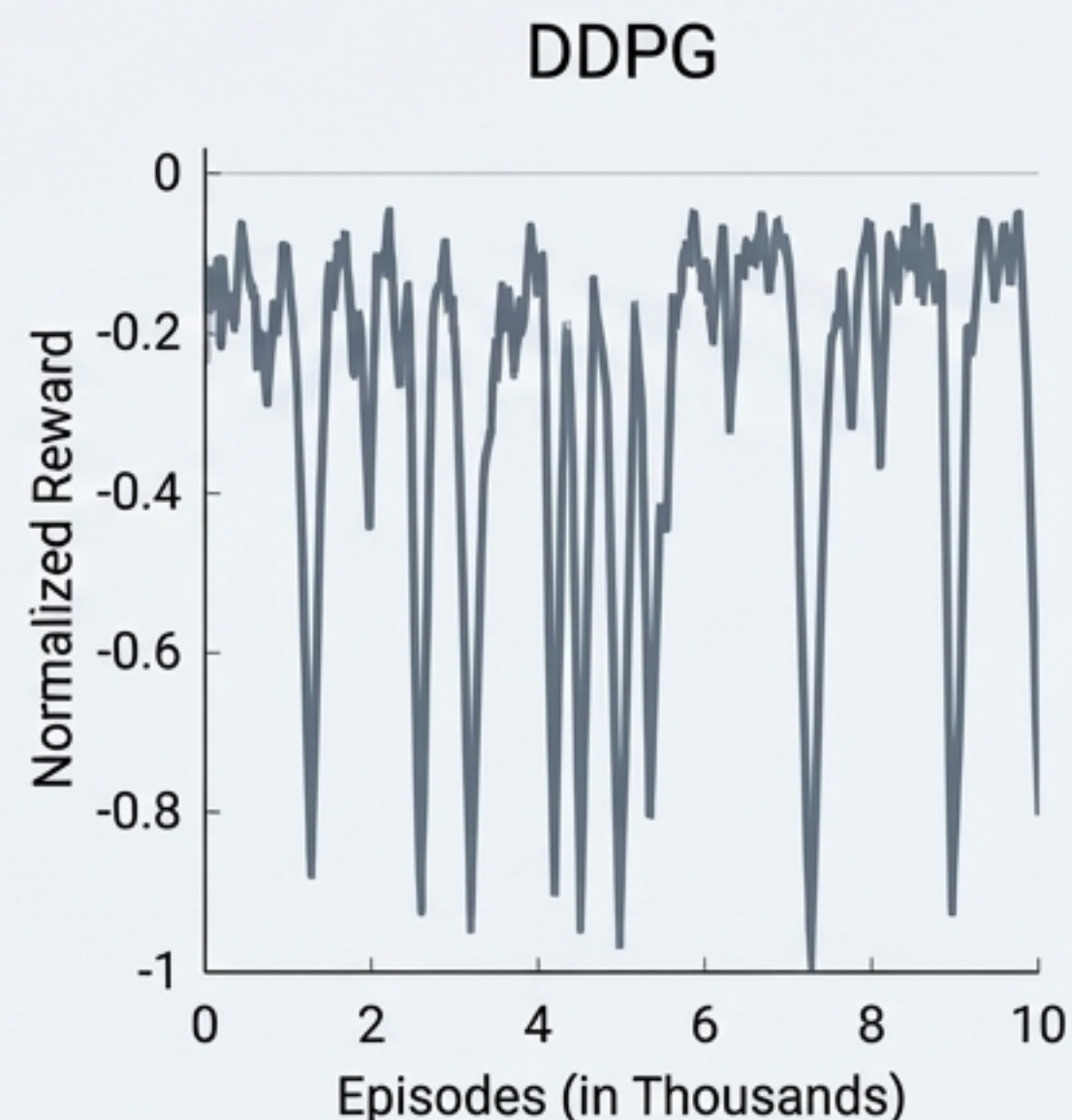
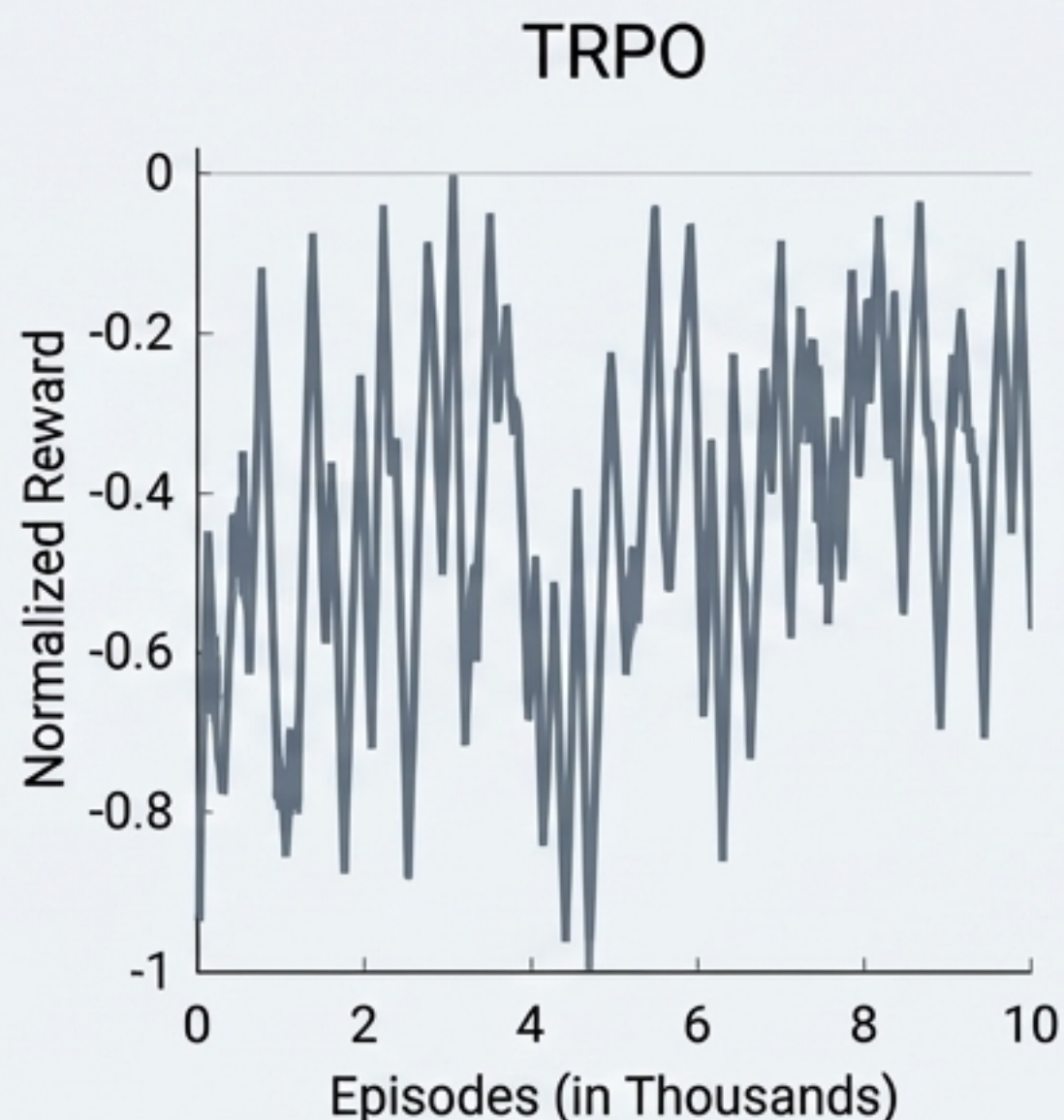
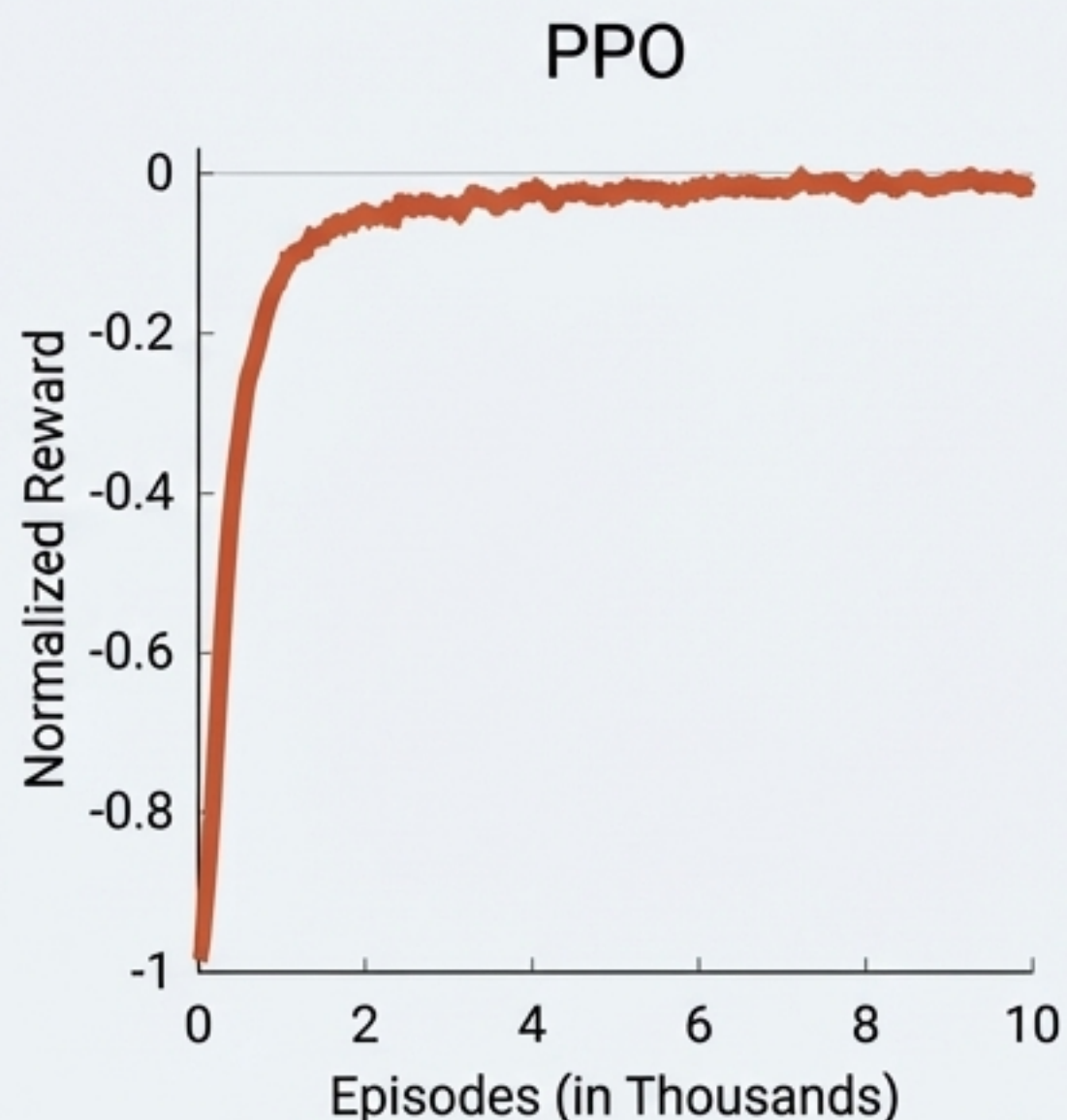
Physics Simulation Steps



Convergence Results: PPO Dominates the Pack

Over the 10-million step training, PPO consistently outperformed TRPO and DDPG in accumulating rewards and achieving stability.

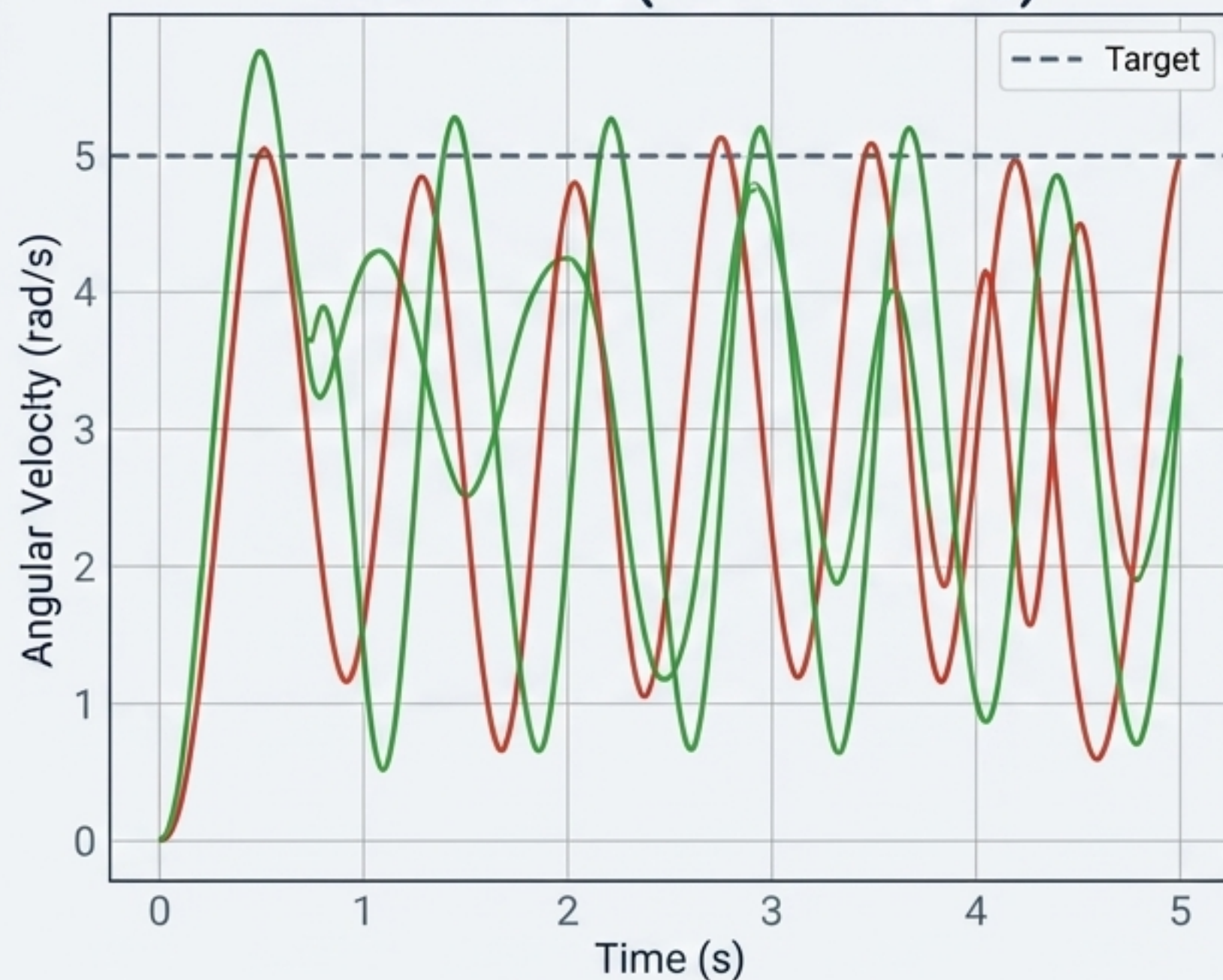
Counterintuitive Insight: Increasing memory size actually decreased convergence across all algorithms by bloating the state space.



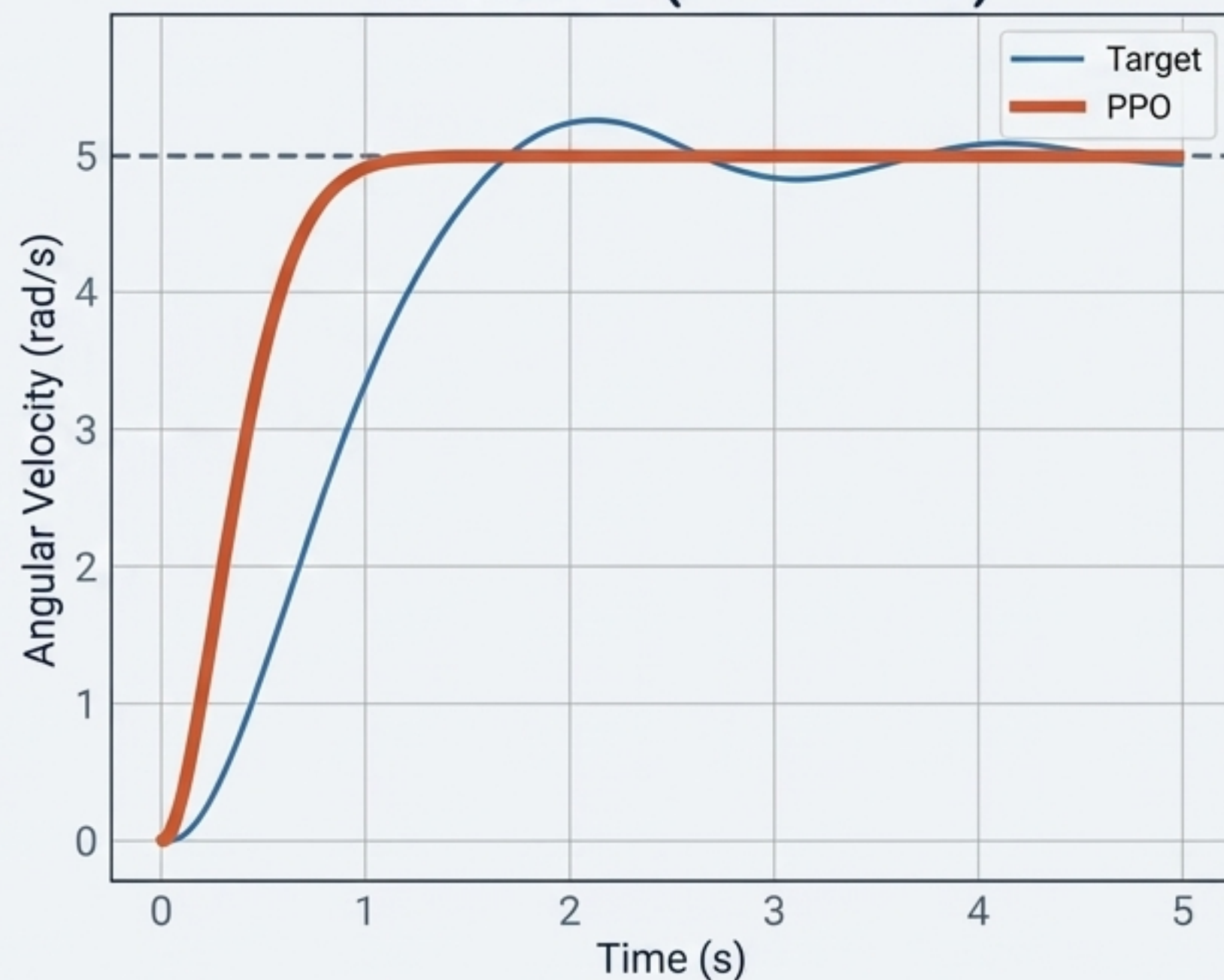
The Showdown: Neural Networks vs. Traditional PID

Evaluated on 1,000 never-before-seen command inputs, TRPO and DDPG suffered from flight-crashing oscillations, while PPO achieved a near-perfect success rate.

The Failures (TRPO & DDPG)



The Winner (PPO vs PID)



By the Numbers: RL Outperforms the Status Quo

PPO exceeded a fully tuned PID controller across almost every critical flight metric.

44%

Decrease in overall error compared to standard PID control.

2.5x

Faster rise time on the pitch axis to reach target velocity.

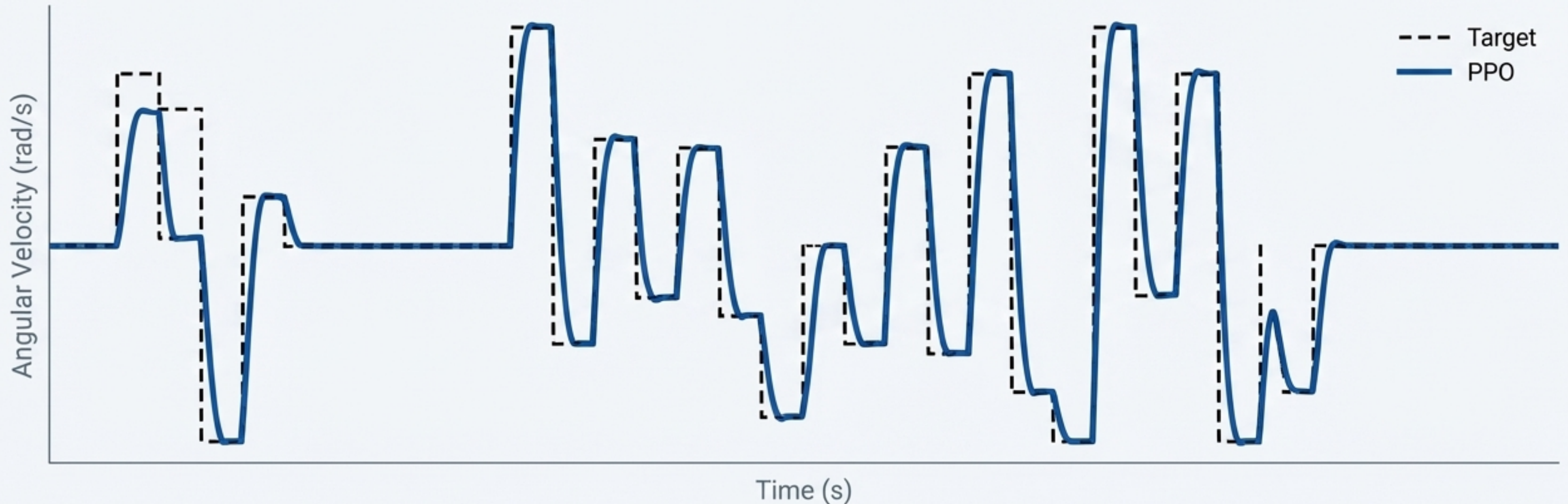
ZERO

Overshoot on roll and yaw axes, achieving perfect stability.

The Continuous Task Surprise

Agents trained strictly on single episodic step-responses performed exceptionally in unpredictable, continuous environments without any retraining.

The Result: A 22% decrease in overall error compared to the PID agent in continuous tracking.



A New Paradigm for Flight Control



Precision: RL is Ready for the Inner Loop

Neural networks can successfully achieve the micro-second precision required for multi-axis attitude control.



Algorithm: PPO is the Definitive Architecture

PPO consistently converges faster and categorically outperforms both traditional PID and competing RL models.



Simulation: Digital Twinning Works

High-fidelity simulated environments like GymFC effectively close the reality gap for eventual hardware deployment.

Escaping the Simulation: The Future Horizon

- 1. Domain Adaptation:**
Training generalized flight dynamic models to instantly adapt to novel aircraft frames.
- 2. Dynamic Reward Engineering:**
Pushing agents to learn complex survivability behaviors, like flying with damaged rotors.
- 3. Real-World Flight:**
Transferring the optimized neural network from the digital twin directly onto physical UAV hardware.

